

Case Study: Library Management System (LMS)

Background

The university wants to build a simple Library Management System.

- Students can borrow and return books.
- Staff can add or remove books.
- The system should follow the principles of OOP (Encapsulation, Inheritance, Polymorphism, Abstraction).

You are assigned as Java developers to implement this system.

Task 1: Encapsulation

Goal: Protect book details using private fields and provide getters/setters.

Instructions:

- Create a Book class with: title, author, isAvailable (default true).
- Make fields private.
- Write getters/setters.
- Write a small main method to test creating a book and checking availability.

Task 2: Inheritance

Goal: Reuse code using inheritance.

Instructions:

- Create a Person class with name and id.
- Create a Student class that extends Person, adding department.
- Add a borrowBook(Book b) method in Student.
- Test borrowing a book (when available vs. not available).

Deliverable:

Demonstrate that Student inherits common properties from Person.

Task 3: Polymorphism

Goal: Use the same reference type (Person) to represent different objects.

Instructions:

- Create a Staff class that extends Person.
- Add a manageBook(Book b, boolean add) method in Staff.
- In main, declare a Person reference pointing to both Student and Staff.
- Show different behaviors when calling methods.

Deliverable:

Show that the same type (Person) can behave differently depending on the object.

Task 4: Abstraction

Goal: Hide implementation details and enforce a contract.

Instructions:

- Create an abstract class `LibraryUser` with: name, abstract method `accessLibrary()`.
- Implement two subclasses: `StudentUser` → 'borrows books', `StaffUser` → 'manages library'.
- Demonstrate abstraction by calling `accessLibrary()` on different users.

Deliverable:

Show that abstract methods enforce implementation in subclasses.

Task 5: Integration (Final Project)

Goal: Combine everything into `LibraryDemo`.

Instructions:

- Create: 2 books, 1 student, 1 staff member.
- Perform actions: borrowing, returning, adding/removing books, accessing library.
- Demonstrate all 4 OOP pillars.